

Localizing Generative Choices in Diffusion Models

a review

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Source: <https://arxiv.org/abs/2604.06052v3>

Attention, May I Have Your Decision?

Localizing Generative Choices in Diffusion Models

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Problem statement

Related work

Methods

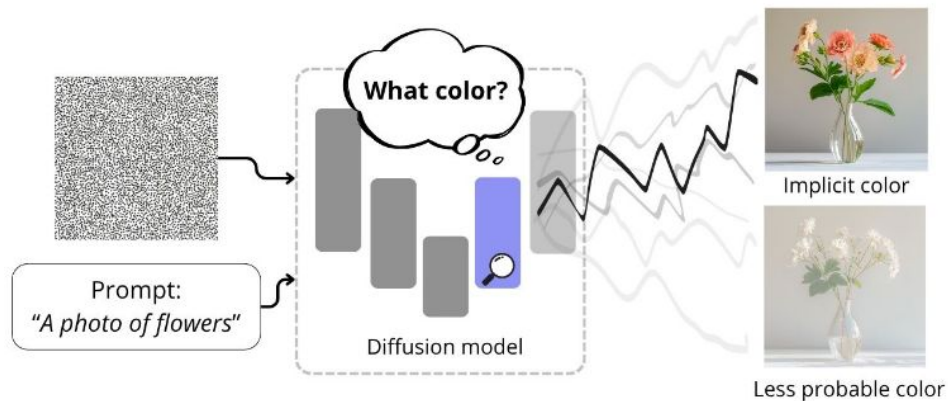
Experimental results

Limitations and future directions

Implicit decisions and bias

Text-to-image diffusion produces high-quality results

Under-specified prompts require from the model to take biased decisions



Explicit prompt: *"A photo of red flowers"*
General prompt: *"A photo of flower"*

Which parts of the model take these decisions?

How can we reduce the bias while preserving image quality?

Problem statement

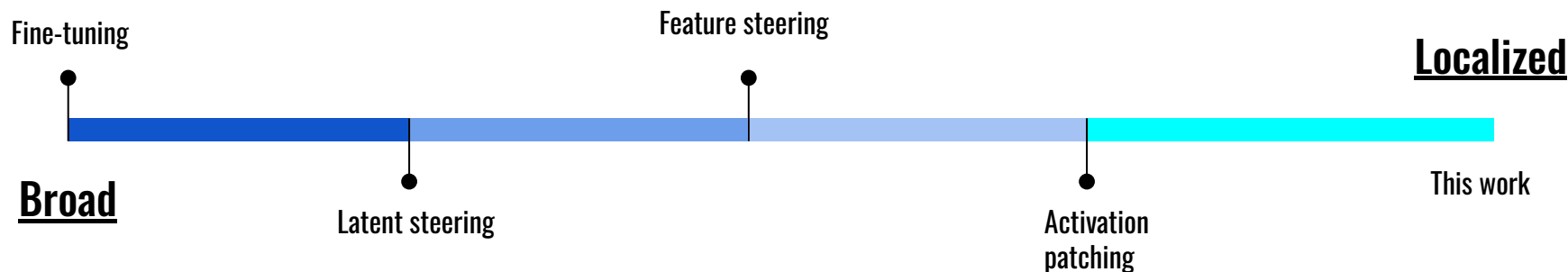
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- **Latent space steering:** adjust hidden directions at inference time ([Li et al. 2024](#)).
- **Fine-tuning approaches:** retrain weights toward a balanced prior.
- **Feature-based steering:** isolate biased features via sparse autoencoders ([Shi et al. 2025](#)).
- **Activation patching:** swap activations between clean/corrupted prompts to find causal layers ([Meng et al. 2022](#)).



Problem statement

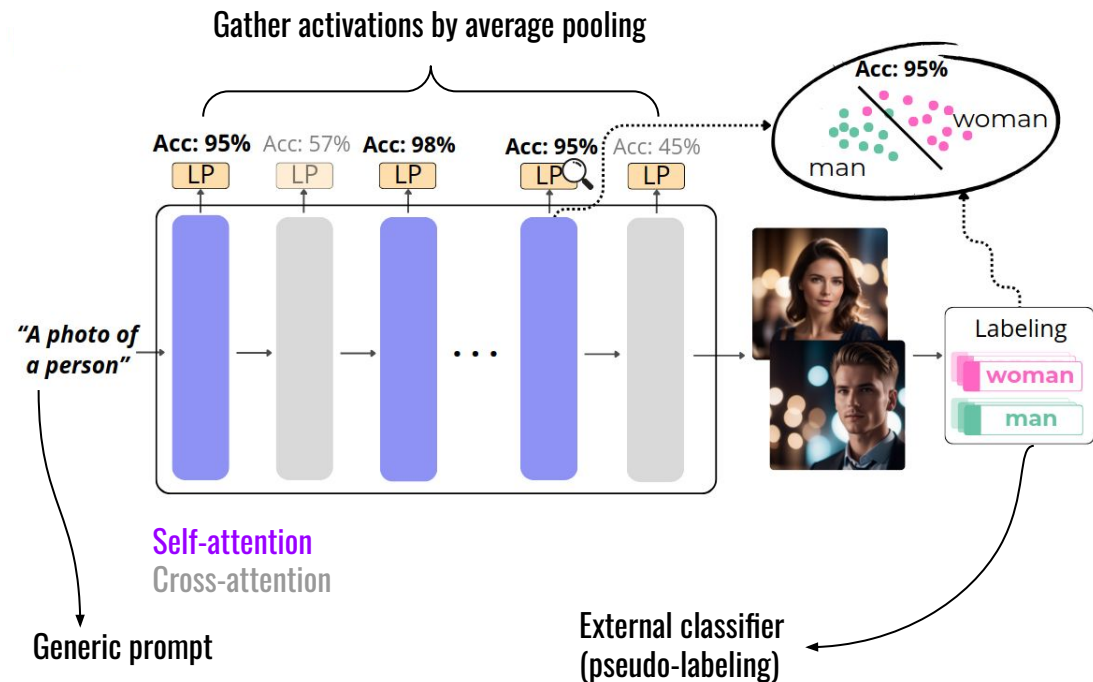
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Implicit Choice-Modification (ICM)



Three stage process

1. Image generation and activation gathering
2. Pseudo-labeling via external classifier
3. Training layer-wise linear probes to quantify discriminability

Result: Locate discriminating layers

Implicit decisions are made in self-attention layers.

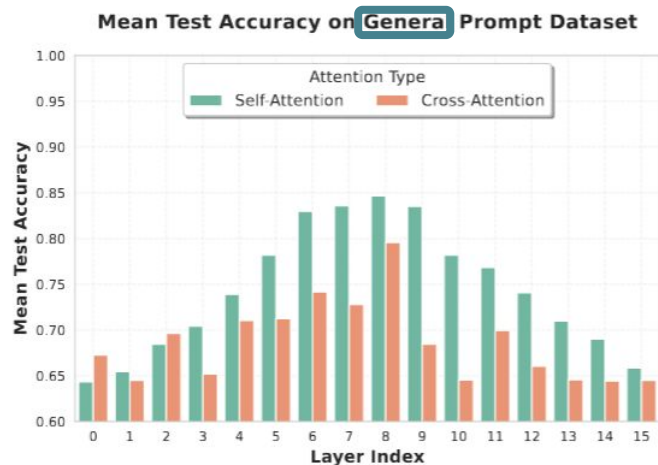
There is no specific token to attend to, thus cross-attention layers perform worse in classification.

Explicit and implicit decisions

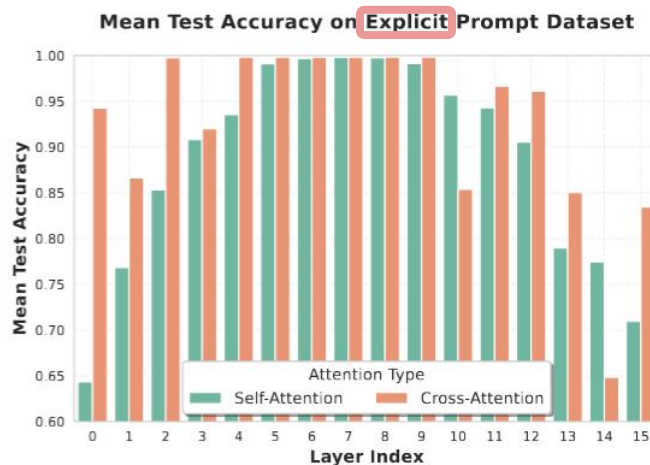
A classifier predicts *gender* based on activations of different attention layers of a UNet.



"A photo of a person"



"A photo of a man"



Decisions for **explicit** prompts are made in cross-attentions layers

Decisions for **implicit** prompts are made in self-attention layers

Now that we know which modules are responsible, how can we approach them?

Activation steering

1. Identify discriminating layers via ICM
2. Given the decision boundary $s_{l,t}$ for layer l and time-step t update activations as

$$H'_{l,t} = H_{l,t} + \alpha s_{l,t}$$

(at inference-time)

Fine-tuning

1. Construct a dataset of triplets $(x, p_{\text{gen}}, p_{\text{spec}})$
 x generated image from prompt p_{spec} but classified under p_{gen}
2. Fine-tune parameters of specific layers

(at training-time)

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Method	bias	image quality	similarity with reference image	alignment with prompt
	FD ↓	FID ↓	CLIP-I ↑	CLIP-T ↑
Original	0.564	120.06	-	0.6155
Finetuning [34]	0.050	161.47	0.8779	0.6095
ICM (Finetuning)	0.535	143.98	0.9187	0.6189
Latent Editing [16]	0.408	166.11	0.8253	0.6005
H-Distribution [26]	0.222	151.68	0.8475	0.6087
Latent Direction [19]	0.305	129.37	0.8058	0.6091
DIFFLENS [35]	0.046	112.83	0.8501	0.6090
ICM (Steering)	0.087	122.08	0.8500	0.6140

Significant bias reduction

Retains image quality

Trade-off

Bias reduction against image quality and prompt alignment

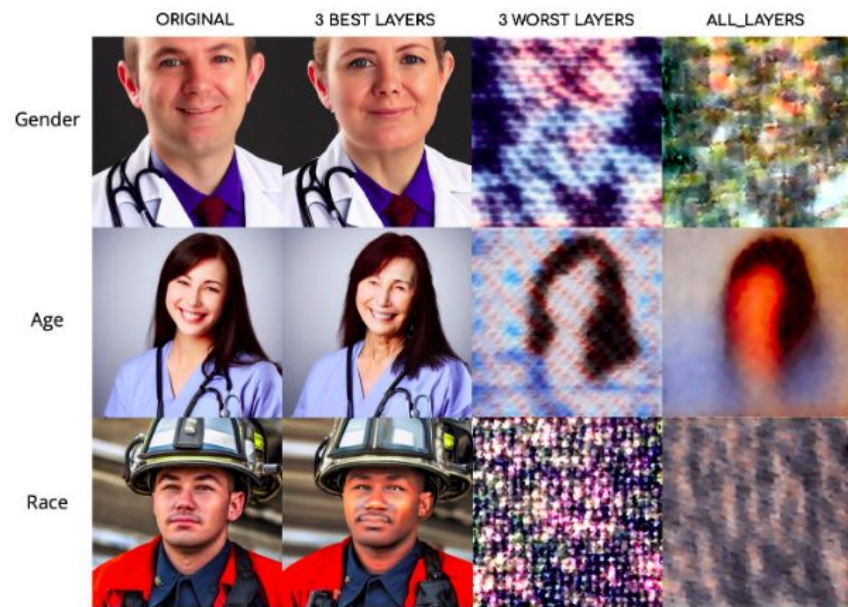
Fairness Discrepancy

prior distribution

predicted attribute
distribution

$$FD := \left\| \bar{p} - \mathbb{E}_{x \sim p_{\theta}(\mathbf{x})}(\mathbf{y}) \right\|_2$$

samples drawn

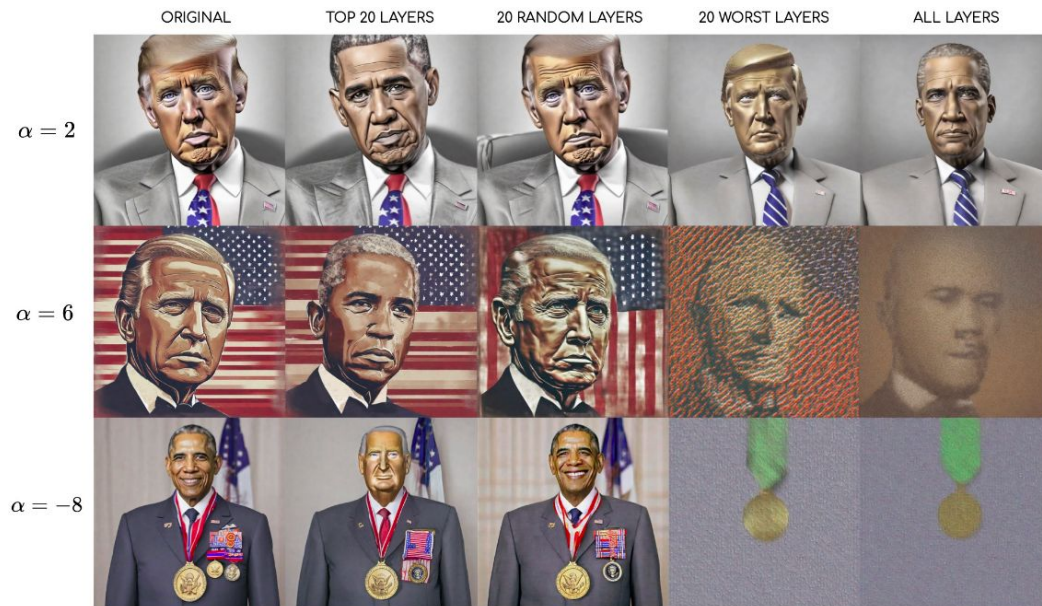


- Implicit decisions are indeed localized
- ICM avoids artifacts
- Average pooling successfully captures activations

Context and background remain the same

Layer selection matters

Scaling up to SDXL



Prompts describing a US president.

Positive class “Barack Obama”, all others are treated as negative.

- Debiasing via ICM scales well
- Preserves context
- Good balance across metrics

Layer Selection	Steering Success (%)	Avg. Confidence	CLIP-I $\uparrow$
Top 20 (Ours)	83.0%	0.917	0.893
Random 20	51.0%	0.838	0.932
All Layers	91.0%	0.879	0.779

Limitations

- The precision of the linear probes is tied to the accuracy and biases of the external classifier.
- Delicate tuning of the steering strength α .
- Capturing activations by average pooling may cause loss of detail.

Future directions

- Multi-factor control: pose, background etc.
- Faster steering for real-time generation.
- Investigate if self-attention is the universal module for implicit decisions in diffusion.